Supplementary Material: Distributed Coherent Optical Computing via Injection-Locked Photonic Networks

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This supplementary material provides additional methodological details and supporting simulation results for the primary manuscript. We describe the empirical criteria used to define the locking-region boundary, the Fourier-domain extraction procedure used to calculate modulation responses, and additional simulations showing how the locking maps depend on linewidth enhancement factor and injected-laser operating threshold. We also include time-domain AM and PM modulation traces to illustrate the extraction of AM→AM, AM→PM, PM→AM, and PM→PM response channels and to visualize nonlinear behavior at large phase-modulation depth.

I. SUPPLEMENTARY METHODS

A. Defining the Locking Range

After plotting the parameter maps of $\Delta I/\bar{I}$ and $\Delta\phi$, a fairly clear transition can be observed in both cases: one side corresponds to a low-value region with comparatively steady behavior, whereas the other corresponds to a region with increased residual oscillation or phase drift. To better identify the approximate range of this transition, we tested a series of threshold values and found that the boundary in the present simulations lies roughly near 0.1 for the intensity metric and 0.1 rad for the phase metric. Accordingly, we use $\Delta I/\bar{I} \leq 0.10$ and $\Delta\phi \leq 0.10$ rad as empirical operational thresholds to help delineate the boundary of the locking region. For the intensity metric, $\Delta I/\bar{I} \leq 0.10$ indicates residual variation below approximately 10% of the mean intensity, which in this work generally separates approximately steady trajectories from states with more pronounced oscillation. For the phase metric, $\Delta\phi \leq 0.10$ rad similarly corresponds to a small residual phase excursion over the analysis window, indicating that the field phase remains approximately stationary. These numerical values are therefore not intended as strict theoretical criteria for injection locking, but rather as empirical thresholds used to more clearly identify and outline the locking-region boundary in the present simulations. This distinction is relevant because prior studies have shown that optically injected lasers can exhibit oscillatory states with bounded phase and average-frequency entrainment near Hopf-mediated locking transitions, even when the intensity is no longer strictly steady-state[1,2]. In that sense, the present thresholds are used to define a conservative practical operating window emphasizing jointly stable amplitude and phase behavior, rather than the broadest possible notion of frequency locking. The cutoff values apply to the intensity and phase metrics themselves, rather than fractions of the plotted color scale. Their effect on the joint-region boundary is evaluated in Sec. IV.A. For the unmodulated maps, we sweep injection ratio from -60 to 0 dB in 1 dB steps and detuning from -30 to 30 GHz in 100 MHz steps. Each trajectory starts from $E(0) = 0$ and $N(0) = N_{\text{th}} = N_0 + 1/(g_0\tau_s)$, with $R_p = 1.15N_{\text{th}}/\tau_n$ and the coupling factor specified in the main text. We integrate the rate equations using fourth-order Runge–Kutta with a 0.2 ps step for 90,000 steps (18 ns), and evaluate the intensity and unwrapped-phase metrics over the final 3,000 samples (0.6 ns). The constant injection field is $E_{\text{inj}} = A_{\text{fr}} 10^{R_{\text{inj,dB}}/20}$, where A_{fr} is the free-running field amplitude from the same model.

B. Fourier-Domain Extraction of Modulation Response

For each operating point $(\Delta\nu, R_{\mathrm{inj}}, f_{\mathrm{mod}})$, the rate equations are first integrated through a transient interval, and the modulation response is extracted from the steady-state measurement window. The simulated injected-laser field is written as

$$E_{\mathrm{injected}}(t) = \mathrm{Re}\{E_{\mathrm{injected}}(t)\} + i\mathrm{Im}\{E_{\mathrm{injected}}(t)\}. \tag{1}$$

The amplitude response channel is defined as the normalized field-amplitude fluctuation,

$$a_s(t) = \frac{|E_{\mathrm{injected}}(t)|}{\langle |E_{\mathrm{injected}}(t)| \rangle} - 1, \tag{2}$$

where $\langle \cdot \rangle$ denotes the time average over the measurement window. The phase response channel is obtained from the optical phase,

$$\phi_s(t) = \mathrm{unwrap}\left[\arg(E_{\mathrm{injected}}(t))\right]. \tag{3}$$

For the modulation-response extraction, the unwrapped phase traces are linearly detrended over the measurement window to remove any residual average phase drift before extracting the modulation component.

The fundamental response amplitude at the imposed modulation frequency is extracted by projecting each input and output waveform onto the sine and cosine components at f_{mod}. Specifically, for a waveform $y(t)$ sampled over the measurement window, the amplitude at f_{mod} is obtained from the quadrature components at angular frequency $2\pi f_{\mathrm{mod}}$. This procedure is equivalent to extracting the Fourier component at the imposed modulation frequency and avoids using broadband spectral features unrelated to the driven response. The resulting input and output amplitudes are denoted as $A_{\mathrm{in}}(f_{\mathrm{mod}})$ and $A_{\mathrm{out}}(f_{\mathrm{mod}})$, respectively.

The response is reported in decibels as

$$H_{\mathrm{dB}}(f_{\mathrm{mod}}) = 20\log_{10}\left[\frac{A_{\mathrm{out}}(f_{\mathrm{mod}})}{A_{\mathrm{in}}(f_{\mathrm{mod}})}\right]. \tag{4}$$

For AM-driven injection, A_{in} is extracted from the normalized remote amplitude modulation. For PM-driven injection, A_{in} is extracted from the unwrapped and linearly detrended remote phase modulation. The same procedure is applied to the corresponding AM and PM output channels to obtain the AM→AM, AM→PM, PM→AM, and PM→PM responses.

II. SUPPLEMENTARY LOCKING-REGION MAPS

The main text shows representative locking maps for the two limiting cases $\alpha = 0$ and $\alpha = 5$. To show the continuous evolution between these two limits, Fig. 1 provides the full set of two-dimensional maps for $\alpha = 0$–5 at threshold = 1.15. The upper panel shows the relative intensity swing $\Delta I/\bar{I}$ together with the locking boundary, while the lower panel shows the corresponding binary locking-region map. These results show that the locking-region boundary evolves gradually with increasing linewidth enhancement factor, including changes in both the extent and asymmetry of the locking region.

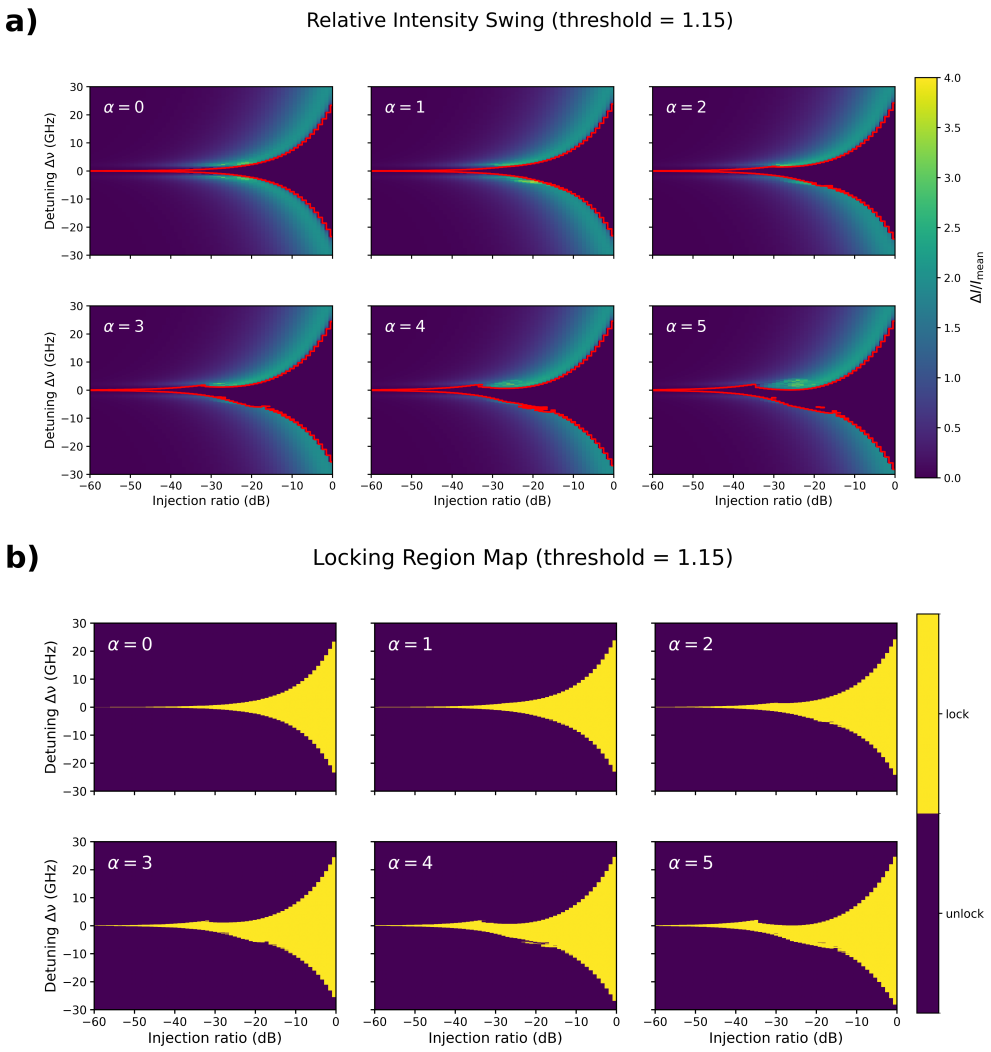


FIG. 1. Supplementary locking-region maps for threshold = 1.15. (a) Relative intensity swing maps for $\alpha = 0$–5. The color scale shows $\Delta I/\bar{I}$, and the red contours indicate the empirical locking-region boundary defined by $\Delta I/\bar{I} \leq 0.10$ and $\Delta\phi \leq 0.10$ rad. (b) Corresponding binary locking-region maps. Yellow regions satisfy both empirical locking criteria, while purple regions do not.

Figure 2 shows the corresponding results for threshold = 1.94. Comparison with Fig. 1 shows that the locking maps depend not only on α, but also on the injected laser operating condition. In particular, changing the threshold or pump-normalized operating point modifies the locking-region extent and boundary shape as a function of injection ratio and detuning.

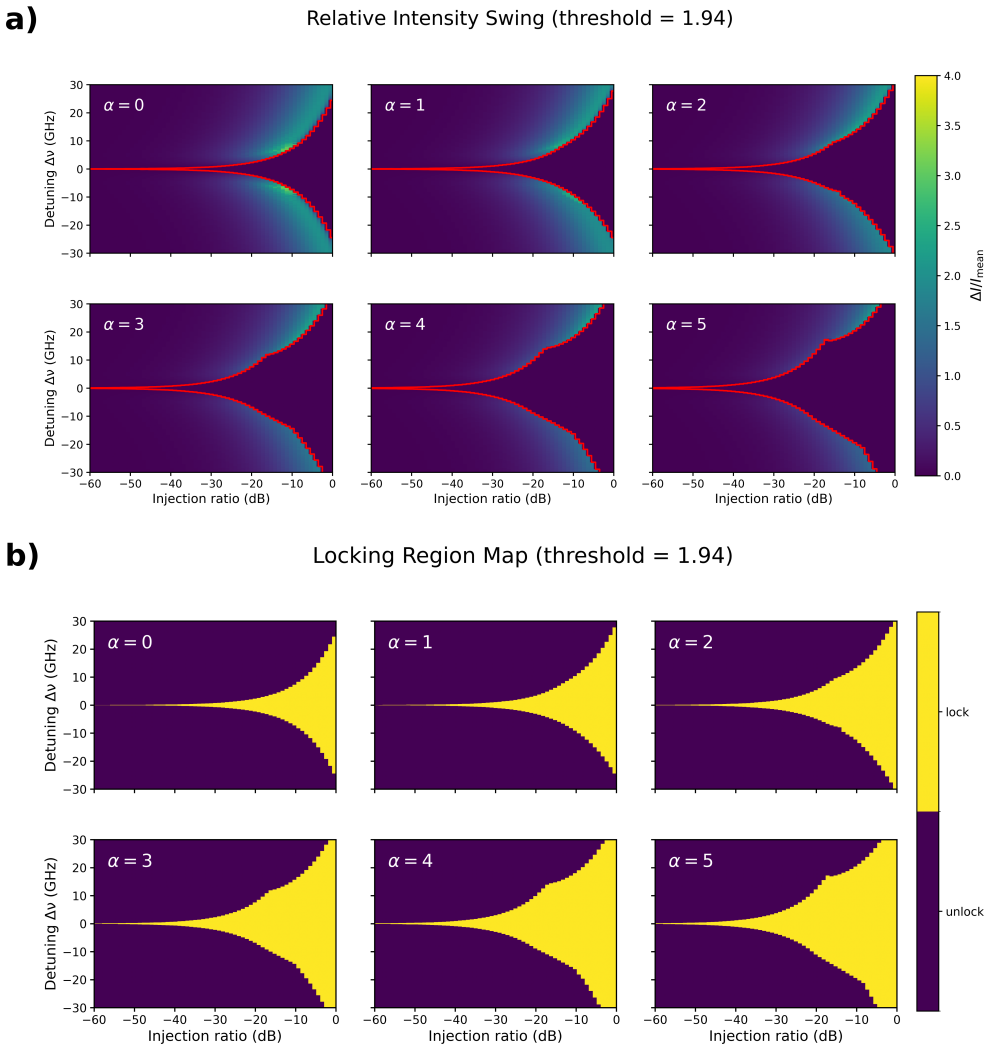


FIG. 2. Supplementary locking-region maps for threshold = 1.94. (a) Relative intensity swing maps with the empirical locking-region boundary. (b) Corresponding binary locking-region maps. Compared with the threshold = 1.15 case, the locking-region boundary and accessible locking range change with the injected laser operating condition.

III. TIME-DOMAIN ILLUSTRATION OF MODULATION CHANNELS

Figure 3 illustrates the time-domain input and output channels used for extracting the modulation response. The traces are calculated at a representative operating point with $\alpha = 5$, $R_{\mathrm{inj}} =$

-50 dB, zero detuning, and $f_{\mathrm{mod}} = 2$ GHz. After a transient interval, the last eight modulation cycles are plotted for modulation depths from 0.60 to 1.00. For AM-driven injection, the input channel is the normalized source-field amplitude modulation, and the output channels are the normalized injected-laser field amplitude and the injected-laser field phase. These channels correspond to the AM→AM and AM→PM responses. For PM-driven injection, the input channel is the source-field phase modulation with depth $m_\phi = 0.6\pi$ to 1.0π, and the output channels are again the injected-laser field amplitude and phase, corresponding to the PM→AM and PM→PM responses. Thus, the figure provides a direct time-domain visualization of the channels used in the Fourier-domain extraction procedure described above.

This figure also provides additional detail for the nonlinear phase-modulation behavior discussed in the main text. In the small-signal regime, increasing modulation depth would primarily scale the response amplitude. However, for PM modulation depths of approximately 0.6π and above, the PM→AM and PM→PM traces show changes in waveform shape and relative amplitude that are not simple proportional rescalings. These time-domain traces therefore help visualize the onset of nonlinear response at large PM depth. For these traces, the dimensionless PM depth $m_\phi = \Delta\phi/\pi$ ranges from 0.6 to 1.0, corresponding to peak phase excursions of 0.6π–1.0π.

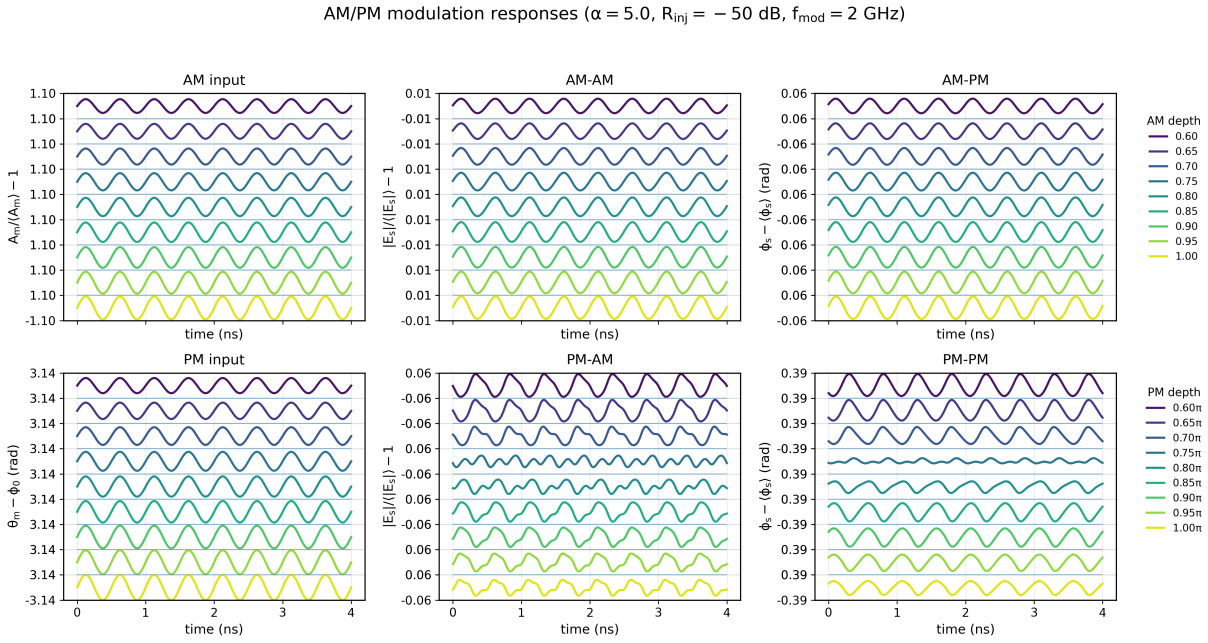


FIG. 3. Time-domain AM and PM modulation responses. The first row shows AM input, AM→AM, and AM→PM channels. The second row shows PM input, PM→AM, and PM→PM channels. The traces illustrate how the four modulation response channels are defined for Fourier-domain extraction and show the nonlinear evolution of the PM-driven response at large phase-modulation depth.

IV. ADDITIONAL SIMULATIONS AND EXPERIMENTAL COMPARISON

A. Sensitivity to the Locking Criteria

Figure 4 compares the joint intensity and phase contours obtained from the same trajectories and observation window. Reducing both cutoffs from 0.10 to 0.01 changes the enclosed area by 0.23% for $\alpha = 0$ and 0.96% for $\alpha = 5$. This weak dependence supports their use for outlining the low-variation region; it does not establish a bifurcation boundary or convergence with observation time. The intensity-only contours may be more sensitive to the cutoff.

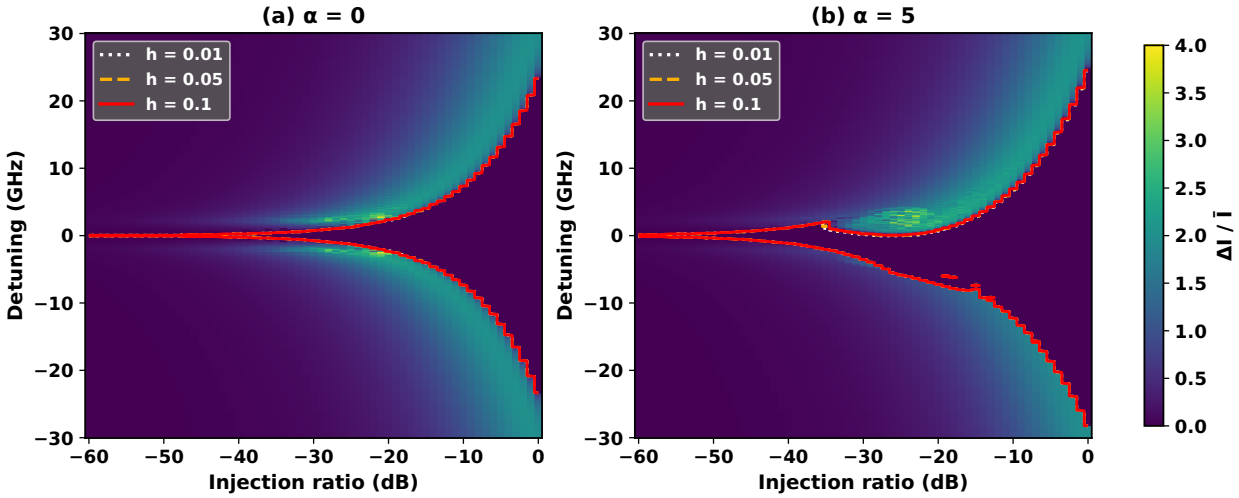


FIG. 4. **Sensitivity of the joint-region contours to the metric cutoffs.** Relative intensity-swing maps for **a)** $\alpha = 0$ and **b)** $\alpha = 5$. Contours satisfy both $\Delta I/\bar{I} \leq h$ and $\Delta\phi \leq h$ rad for $h = 0.01$, 0.05, and 0.10.

B. Responses at Small Modulation Depths

Figure 5 supplements main-text Fig. 5 with responses at 1%, 5%, and 10% modulation depth, extracted from the same frequency-response calculations.

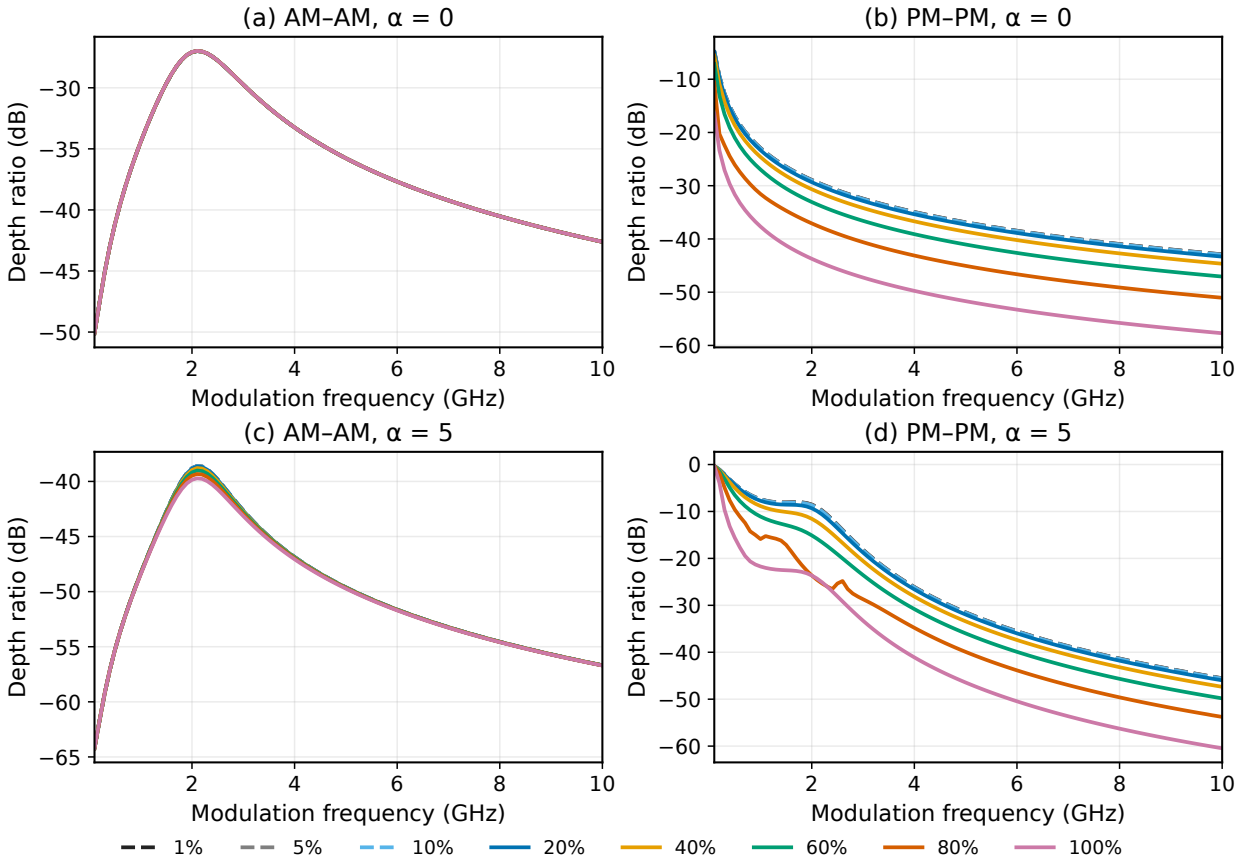


FIG. 5. **Modulation responses over an extended selection of depths.** AM→AM and PM→PM fundamental responses at $R_{\mathrm{inj}} = -50$ dB and zero detuning for **a,b)** $\alpha = 0$ and **c,d)** $\alpha = 5$. Curves correspond to depths of 1%, 5%, 10%, 20%, 40%, 60%, 80%, and 100%.

C. Comparison with Fiber Measurements

The unmodulated measurements in Fig. 6 exhibit a detuning-dependent locking boundary and residual oscillation features outside the low-response region. Under modulated injection, the measured response is also suppressed within part of the locked region. We compare these features with the simulated intensity spectrum and fundamental amplitude response, respectively.

To account for PM accompanying the AM produced by the experimental MZM, we use

$$E_{\mathrm{inj}}(t) = A_{\mathrm{inj},0}[1 + 0.1\sin(2\pi f_{\mathrm{mod}} t)]\exp[i\,0.10\sin(2\pi f_{\mathrm{mod}} t)]. \tag{5}$$

The field-amplitude modulation depth is 0.1, and the peak PM is 0.10 rad, in phase with the AM drive. Using these parameters, the simulation results show behavior similar to the experimental results, with a deep trench marking the lowest transfer-function response in a similar detuning range. The PM amplitude is an effective model parameter rather than an independently measured EOM chirp.

We use $\alpha = 2$ and a normalized pump setting of 1.944 to represent injected-laser output near 0 dBm. Matching the external experimental injection ratio of -15 dB to the effective simulated ratio of -40 dB gives an estimated power-coupling offset of -25 dB. This estimate depends on the model calibration and does not independently determine κ.

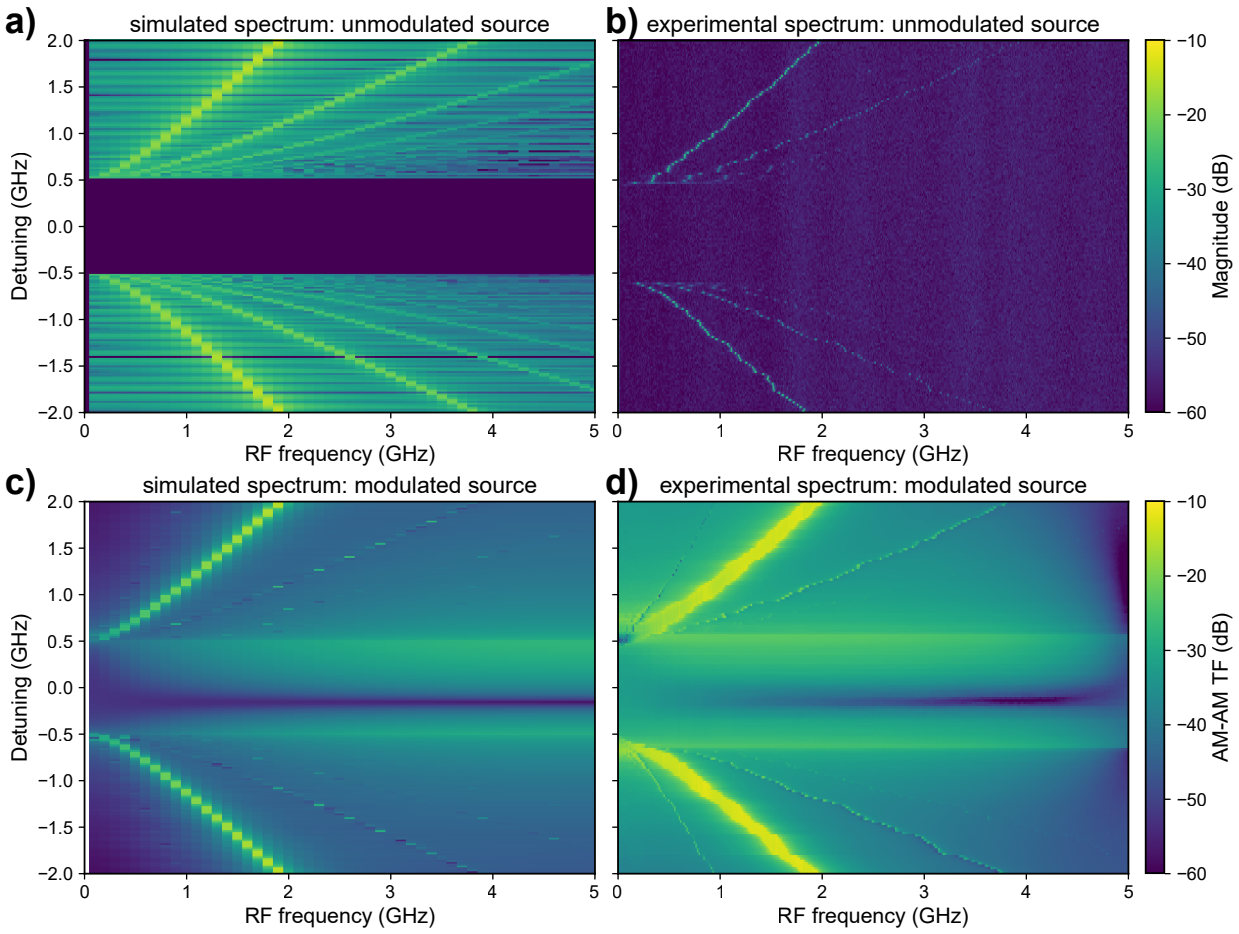


FIG. 6. **Comparison of fiber measurements and simulation.** Unmodulated injection: **a)** simulated injected-laser intensity spectrum and **b)** spectrum measured in SVA spectrum mode. Modulated injection: **c)** simulated amplitude response and **d)** response measured in VNA mode after reference subtraction and optical-power normalization. The simulated fractional field-amplitude response at the drive frequency is divided by the input AM depth. Horizontal and vertical axes show RF frequency and optical detuning, respectively. All panels use color limits of -60 to -10 dB, with spectral normalization in the upper row and transfer-function normalization in the lower row.

D. Complex-Field Readout and Facet Reflection

For Fig. 7, we retain the modulation sequence and timing of main-text Fig. 6. Independent random generators with seeds 4 and 8 produce the remote and local inputs, respectively. The zero-order-held waveforms pass through a 20 GHz Gaussian FIR filter and are sampled at 64 points per encoded symbol. The active sequence contains 200 encoded symbols, of which the first 100 are displayed.

We represent the field incident on the local remodulator as the sum of the rate-equation output and a coherent reflected component:

$$\begin{aligned} E_{\mathrm{out}}(t) &= E_{\mathrm{injected}}(t) + \sqrt{R_f}E_{\mathrm{inj}}(t), \\ E_{\mathrm{injected}}^{(\mathrm{modulated})}(t) &= \frac{E_{\mathrm{out}}(t)}{\langle E_{\mathrm{injected}}\rangle_{\mathrm{CW}}}\, y(t). \end{aligned} \tag{6}$$

The power reflectivity is $R_f = 0$ or 0.30, giving a reflected-field coefficient $\sqrt{R_f}$. The injected field is $E_{\mathrm{inj}} = A_{\mathrm{fr}} 10^{R_{\mathrm{inj,dB}}/20}[1 + x(t)]$, and the signed computational data field is $E_{\mathrm{remote}}^{(\mathrm{modulated})} = x(t)$. We normalize both reflection cases by the complex mean of the no-reflection field over 10–19 ns, before data modulation. This fixed calibration is independent of the input data. Both fields are expressed in effective output-plane units with unit intracavity-to-output conversion. The laser parameters are held fixed in this comparison; quantitative predictions for a particular device require calibration of facet coupling and output conversion.

The balanced-detector output and its accumulation are evaluated at every time sample:

$$D_j = 2\,\mathrm{Re}\left\{ x_j \left[E_{\mathrm{injected},j}^{(\mathrm{modulated})} \right]^* \right\}, \qquad A_n = \frac{1}{64} \sum_{j=1}^{n} D_j, \tag{7}$$

where sampling begins at the onset of data modulation at 20 ns. At symbol boundaries, A_n equals the sum of the preceding symbol-averaged products. The ideal reference is $D_j^{\mathrm{ideal}} = 2x_j y_j$, evaluated using the same filtered inputs. The display transformation $(1 + \mathrm{Re}\,E)/2$ in main-text Fig. 6b is not used in the readout calculation. A reflected contribution can reinforce or partly cancel the existing error for a given input; the statistical comparison below assesses its effect across multiple inputs.

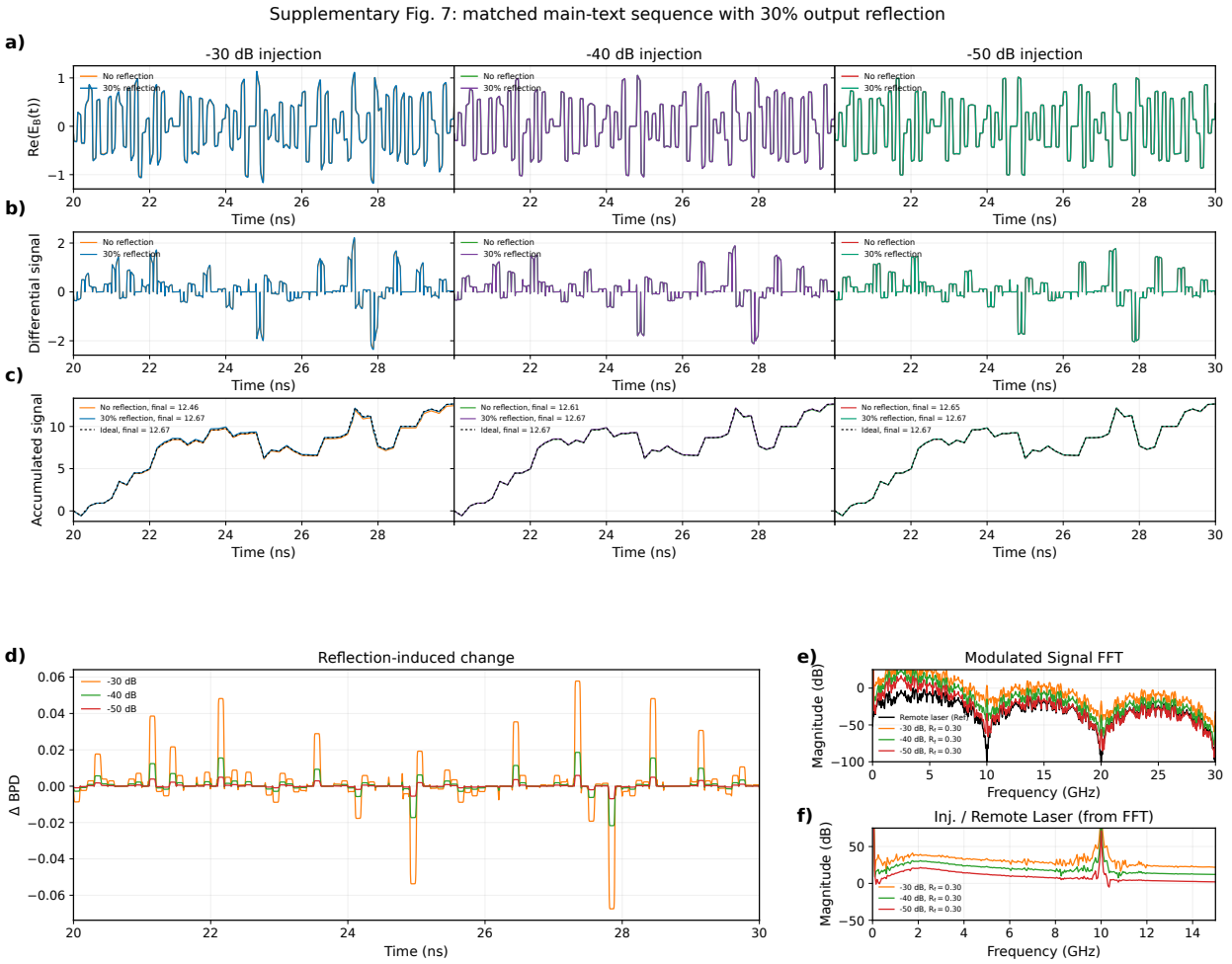


FIG. 7. **Effect of output reflection at $\alpha = 0$. a–c)** Remodulated injected-laser fields, BPD signals, and accumulated outputs over 20–30 ns at injection ratios of -30, -40, and -50 dB. Contrasting solid curves distinguish $R_f = 0$ and 0.30; dotted curves in (c) show the ideal reference, with values at 30 ns listed in the legends. **d)** Change in BPD output due to reflection. **e)** Spectra before local remodulation, including the injected-laser carrier, residual modulation, and reflected field, compared with the remote spectrum. **f)** Injected-to-remote spectral ratios from (e).

E. Accumulated-Error Statistics over 1000 Inputs

We evaluate $Q = 1000$ independent remote/local input pairs with 1024 active encoded symbols per vector. The same pairs are used for all combinations of $\alpha = 0, 2, 5$, injection ratios of -50, -40, and -30 dB, and $R_f = 0$ or 0.30, at zero detuning. Remote and local vectors are generated independently. The inputs retain the 100% field-modulation depth, eight-level magnitudes, antipodal encoding, filtering, sampling, and 20 ns CW settling interval used above. Results at length L are evaluated from the first L symbols of each trajectory.

For input pair q, we define the accumulated error and ensemble root-mean-square error as

$$\begin{aligned} \varepsilon_q(L) &= A_{\mathrm{OIL},q}(L) - A_{\mathrm{ideal},q}(L), \\ A_{\mathrm{OIL},q}(L) &= \frac{1}{64}\sum_{j=1}^{64L} D_{q,j}, \qquad \mathrm{RMSE}(L) = \sqrt{\frac{1}{Q}\sum_q \varepsilon_q(L)^2}. \end{aligned} \tag{8}$$

Relative to the ideal readout defined in Sec. IV.D, these errors isolate the contributions of OIL and reflection. They exclude the difference between filtered-waveform products and the original digital dot product.

The final errors in Fig. 8 are expressed as percentages of a common ideal-output RMS:

$$S = \sqrt{\frac{1}{Q}\sum_q A_{\mathrm{ideal},q}(1024)^2}, \qquad p_q = 100\frac{\varepsilon_q(1024)}{S}. \tag{9}$$

Here S is the RMS of the 1000 ideal accumulated outputs. Using this common denominator for every parameter combination preserves the shape of each error distribution and avoids division by individual ideal outputs near zero. The percentage is therefore relative to the ensemble RMS, rather than each input's ideal result. Gaussian fits characterize the observed distributions without assuming Gaussian statistics.

To compare accuracy across accumulation lengths, Fig. 9 reports the normalized RMSE and Pearson correlation:

$$\begin{aligned} \mathrm{NRMSE}(L) &= \sqrt{\frac{\sum_q \varepsilon_q(L)^2}{\sum_q A_{\mathrm{ideal},q}(L)^2}}, \\ r_{\mathrm{P}}(L) &= \mathrm{corr}_q[A_{\mathrm{OIL},q}(L), A_{\mathrm{ideal},q}(L)]. \end{aligned} \tag{10}$$

NRMSE is plotted as a percentage of the ideal-output RMS at each L. Correlation measures linear association across input pairs and is insensitive to an additive offset or a positive overall gain; it therefore complements, but does not replace, the error metric.

The dependence on α follows from amplitude–phase coupling and the phase sensitivity of coherent detection. Writing the calibrated field without reflection as $E_{\mathrm{injected}}/\langle E_{\mathrm{injected}}\rangle_{\mathrm{CW}} = (1+a_\alpha)e^{i\phi_\alpha}$, we obtain

$$\delta D_\alpha = 2xy[(1+a_\alpha)\cos\phi_\alpha - 1] \simeq 2xy\left(a_\alpha - \frac{\phi_\alpha^2}{2}\right). \tag{11}$$

The residual amplitude and phase fluctuations are denoted by a_α and ϕ_α, respectively. Near the calibrated operating point, amplitude errors enter at first order, while phase errors enter at second order. Amplitude–phase coupling[3] can therefore change the relative contributions of these terms to the detected error. In the simulated cases, nonzero α reduces RMSE at -50 and -40 dB, whereas larger errors occur at -30 dB, particularly for $\alpha = 5$. This expansion describes the small-residual sensitivity; the reported results use the full nonlinear rate equations and exact readout.

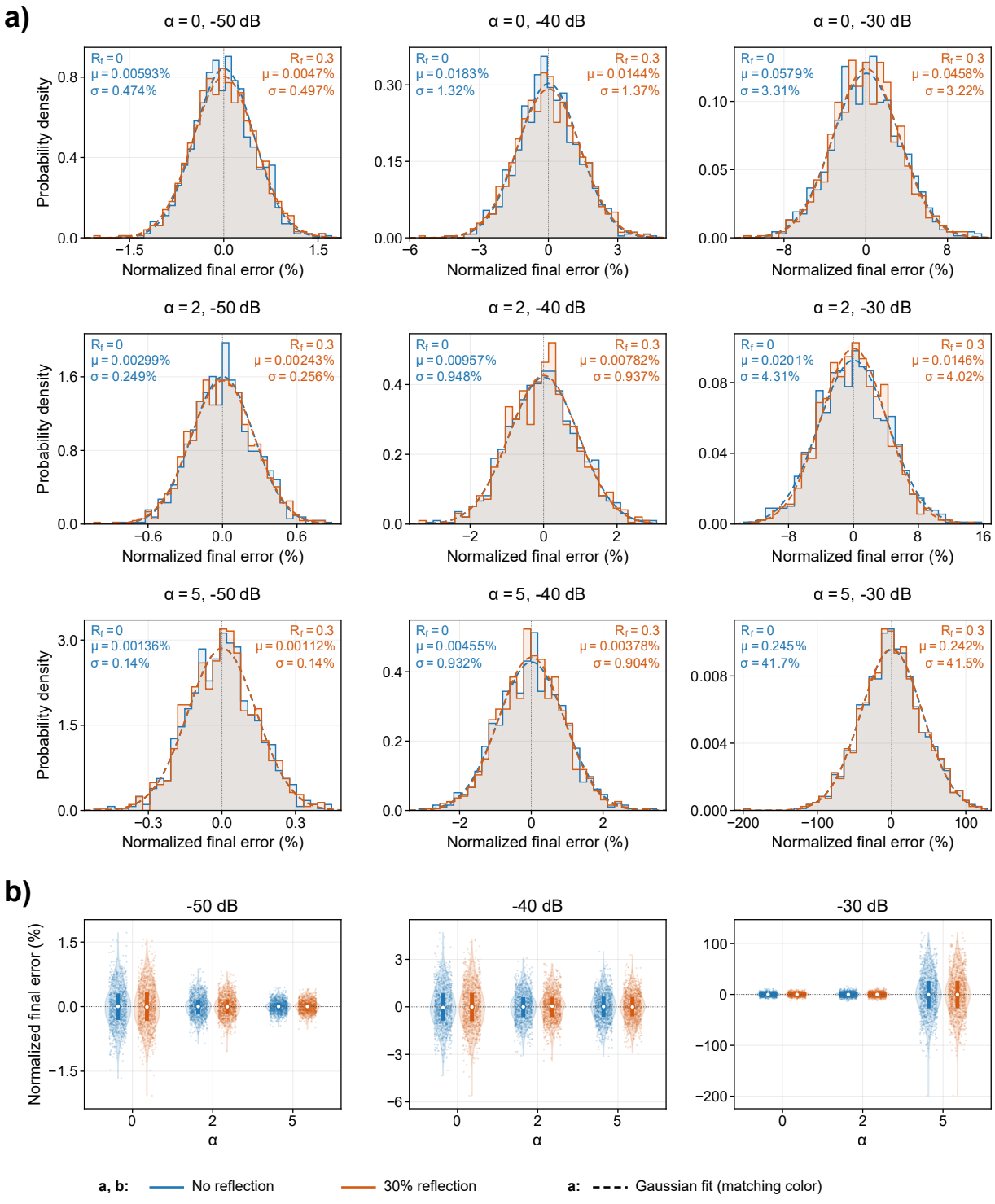


FIG. 8. **Final accumulated-error distributions for 1000 input pairs at $L = 1024$.** Errors are normalized according to Eq. (9). **a)** Histograms with Gaussian fits (dashed curves); fitted means μ and standard deviations σ are given in percentage points. Rows correspond to $\alpha = 0, 2, 5$. **b)** Error distributions versus α, with individual inputs shown as dots, density as violins, interquartile ranges as thick bars, and medians as white dots. Columns correspond to injection ratios of -50, -40, and -30 dB. Blue and orange denote $R_f = 0$ and 0.30, respectively.

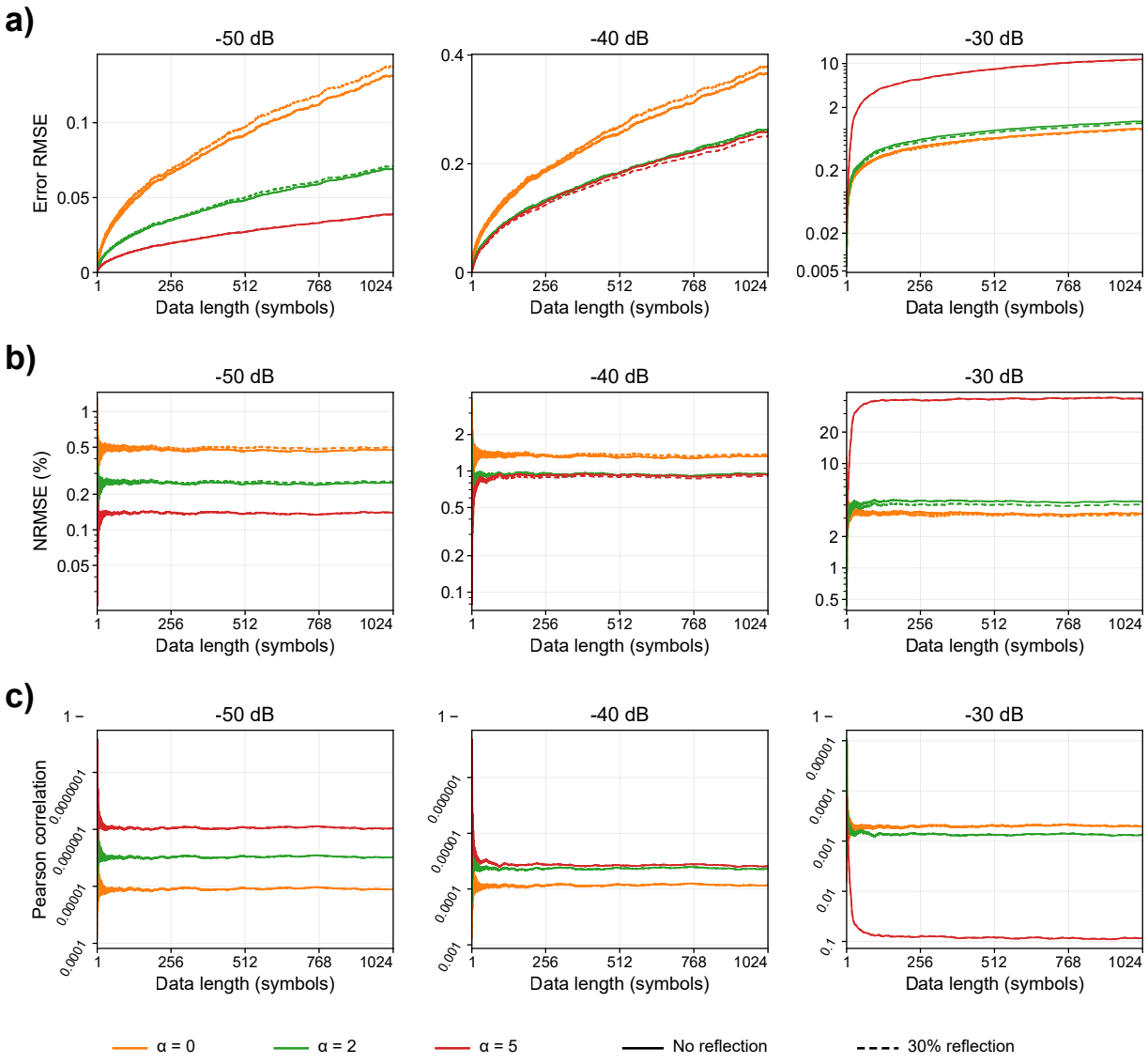


FIG. 9. **Accumulation accuracy versus encoded sequence length. a)** RMSE, **b)** NRMSE (%), and **c)** Pearson correlation over the first $L = 1$–1024 symbols of the inputs in Fig. 8. Columns correspond to injection ratios of -50, -40, and -30 dB; orange, green, and red denote $\alpha = 0, 2, 5$, respectively. Solid and dashed curves denote $R_f = 0$ and 0.30. Each subplot uses its own vertical range. RMSE is plotted on linear axes at -50 and -40 dB and a logarithmic axis at -30 dB; NRMSE uses logarithmic axes. In (c), tick labels express correlation as $1 - r_{\mathrm{P}}$ beneath a $1-$ prefix, with logarithmic spacing in $1 - r_{\mathrm{P}}$.

V. SHORT-REACH IMPLEMENTATION AND STABILIZATION

For the short-reach links considered here, practical operation requires control of both the laser operating point and the relative phase and polarization at the detector. The following measures outline a possible implementation.

The allowable frequency drift depends on the operating detuning and the range over which locking and acceptable modulation suppression coexist. It cannot be inferred from telecom-grade laser specifications alone. The static maps identify candidate operating regions but do not determine a quantitative drift tolerance.

Combined OIL–OPLL systems use optical injection to suppress rapid relative-phase fluctuations and electronic feedback to compensate slower drift, extending the system hold-in range[4]. In the proposed implementation, a filtered carrier-selective monitor or dedicated pilot would provide an error signal for PID control of the injected-laser current or temperature. This avoids relying on the data-bearing sidebands as a locking reference. A separate phase servo before interference would compensate differential optical-path fluctuations; it would not replace control of the laser operating point. Loop bandwidths and residual errors under data modulation require experimental validation.

Polarization-maintaining fiber with aligned launch axes, or active polarization tracking, would preserve injection coupling and polarization overlap at the detector. Acoustic and thermal path-length fluctuations would be addressed through compact routing, environmental isolation, and phase feedback within the control bandwidth. Dispersion can be reduced by shortening the link or using compatible 1310 nm devices near the zero-dispersion wavelength of conventional single-mode fiber[5], with the laser and link parameters recalibrated for that wavelength. Where the power budget permits, omitting in-line amplification also avoids amplifier-added noise. Residual laser noise, loss, and fluctuations beyond the control bandwidth must be characterized for the chosen link; the present calculations do not establish a maximum transmission distance or a complete network-noise budget.

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